Joker of Academics

Did You Even Fucking Reply to Your Advisor Today?

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Abstract

This study innovatively introduces the **nonlinear dynamics framework** into the field of graduate mental health research, focusing on the unexplored correlation between graduate students' submissive reply behaviors (e.g., "Okay, professor", "Received", "Understood") in academic communication with supervisors and academic anxiety. A 12-month longitudinal cohort study was conducted among 317 postgraduate students from comprehensive universities, science and engineering universities, and medical universities in eastern, central, and western China. We quantified five core indicators of submissive replies: reply frequency, lexical category, response latency, periodic fluctuation, and emotional valence of wording, combined with the Generalized Anxiety Disorder-7 (GAD-7) scale, Academic Anxiety Scale for Graduate Students (AAGS), and nonlinear dynamic modeling methods (including phase space reconstruction, Lyapunov exponent calculation, and bifurcation analysis).

Results demonstrated that: (1) Submissive reply features and graduate anxiety present a **significant nonlinear dynamic relationship** rather than a simple linear correlation, with obvious chaotic characteristics and periodic bifurcation phenomena; (2) High-frequency, single-type, and rigid submissive replies (e.g., repetitive "Received" without emotional information) are strong predictive signals of moderate-to-severe academic anxiety (AUC = 0.876, sensitivity = 82.3%, specificity = 81.5%); (3) Nonlinear dynamic models based on reply behavioral indicators achieve high-precision prediction of anxiety changes (R^2 = 0.892), outperforming traditional linear regression models by 37.4%; (4) The periodic mutation of reply behaviors can serve as an **early warning marker** for anxiety deterioration 4–6 weeks in advance.

This study breaks through the limitations of traditional mental health research relying on self-report scales, pioneers the computational quantification of micro-interactive behaviors in supervisor-student relationships, and provides an objective, non-invasive, real-time prediction method for identifying high-anxiety graduate students. The findings have important theoretical value for nonlinear psychological dynamics and practical significance for the early intervention of graduate mental health crises.

Keywords: Submissive Reply Behaviors; Graduate Academic Anxiety; Nonlinear Dynamics; Supervisor-Student Communication; Computational Psychology; Longitudinal Prediction; Chaotic System

1. Introduction

1.1 Background and Research Significance

The mental health crisis of graduate students has become a global public health and higher education management problem (Xiao et al., 2024; Liu et al., 2023). Recent epidemiological surveys show that the detection rate of anxiety symptoms among Chinese graduate students is as high as 42.7%, of which 15.3% reach moderate to severe anxiety levels, far higher than the same-age non-graduate population (National Health Commission of China, 2025). Academic anxiety, as a specific type of anxiety triggered by research tasks, supervisor pressure, academic performance, and graduation pressure, has become the core risk factor leading to graduate students' emotional exhaustion, academic burnout, self-harm, and even suicidal behaviors (Zhang and Wang, 2024).

Existing research on graduate anxiety mostly focuses on macro-stressors (e.g., supervisor style, academic pressure, interpersonal relationships) and relies on self-report psychological scales for cross-sectional research (Li et al., 2023; Chen et al., 2022). However, these methods have obvious limitations:

(1) Self-report scales are susceptible to subjective bias, social desirability effect, and recall bias, leading to low accuracy of anxiety assessment;

(2) Cross-sectional research cannot capture the dynamic evolution process of anxiety;

(3) Macro-stressor research ignores the **micro-behavioral markers** hidden in daily supervisor-student interactions, which are more sensitive and predictive of psychological changes.

Supervisor-student communication is the most frequent and core social interaction scenario in graduate study (Zhao et al., 2024). In online and offline communication, graduate students' submissive replies to supervisors' instructions (e.g., "Okay, professor", "Received", "Good", "Understood professor") are not only linguistic responses but also **external behavioral manifestations of internal psychological state, power cognition, and emotional pressure**. In real educational practice, we observe that students with different anxiety levels show significant differences in reply frequency, wording selection, and temporal regularity: students with mild anxiety tend to use flexible, diverse, and emotionally appropriate replies, while students with severe anxiety often show rigid, repetitive, high-frequency, and overly cautious reply behaviors.

However, the academic community has not yet conducted systematic quantitative research on this phenomenon, and no study has explored the **nonlinear dynamic relationship** between submissive reply behaviors and anxiety. Nonlinear dynamics, as a mathematical tool to study complex systems with chaotic, periodic, and bifurcation characteristics, is highly consistent with the evolutionary law of psychological systems: human mental state is a typical nonlinear dynamic system, which is affected by multiple factors and presents non-single, non-straightforward changes (Guastello et al., 2021; Haken and Schuz-Knol, 2023).

Against this background, this study puts forward a pioneering scientific hypothesis: **graduate students' submissive reply behaviors to supervisors are observable nonlinear dynamic markers of academic anxiety, and their frequency, type, periodicity, and other characteristics can realize high-precision prediction of anxiety levels and dynamic evolution trends through nonlinear dynamic modeling.**

This study has three innovative contributions:

(1) For the first time, it defines and quantifies the submissive reply behavior system in supervisor-student communication, filling the research gap in micro-behavioral markers of graduate anxiety;

(2) It introduces nonlinear dynamics into the field of graduate mental health, revealing the chaotic and periodic evolutionary law between interactive behaviors and anxiety;

(3) It constructs an objective, non-invasive, real-time anxiety prediction model based on behavioral data, providing a new technical path for precise early warning and intervention of graduate mental health crises.

1.2 Research Questions and Hypotheses

1.2.1 Core Research Questions

1. What are the quantitative characteristics (frequency, type, latency, periodicity, valence) of graduate students' submissive reply behaviors in supervisor-student communication?

2. What is the nonlinear dynamic relationship between submissive reply features and graduate academic anxiety? Is there chaotic, periodic, or bifurcation characteristics?

3. Can submissive reply behavioral indicators achieve high-precision prediction of graduate anxiety levels and dynamic changes? What is the predictive efficiency and early warning time window?

4. What psychological mechanisms underlie the correlation between submissive reply behaviors

and anxiety?

1.2.2 Research Hypotheses

1. H1: Submissive reply behaviors and graduate academic anxiety form a nonlinear dynamic system with positive Lyapunov exponents (chaotic characteristics), rejecting the linear correlation hypothesis.

2. H2: High frequency, single type, short latency, rigid periodicity, and neutral-negative valence of submissive replies are positively correlated with anxiety levels, and can predict anxiety deterioration in advance.

3. H3: Nonlinear dynamic models based on submissive reply features have significantly higher prediction accuracy than traditional linear models (logistic regression, linear regression).

4. H4: Psychological mechanisms such as **power distance cognition, impression management motivation, emotional suppression, and threat sensitivity** mediate the nonlinear relationship between submissive replies and anxiety.

1.3 Theoretical Framework

This study is supported by four core psychological and mathematical theories, forming a complete theoretical framework:

1.3.1 Nonlinear Dynamic Psychology Theory

Nonlinear dynamic psychology (also known as dynamic systems psychology) holds that human psychological and behavioral systems are complex dynamic systems composed of multiple interacting variables, which evolve over time and present nonlinear characteristics such as self-organization, bifurcation, chaos, and attractors (Guastello, 2019; Schank, 2020). Unlike linear systems, small changes in input variables (e.g., a slight increase in supervisor pressure) may lead to sharp changes in output variables (e.g., rapid deterioration of anxiety), which is highly consistent with the sudden onset and fluctuating characteristics of graduate anxiety.

1.3.2 Power Distance Theory (Hofstede, 2001)

In the context of Chinese higher education, the supervisor-student relationship has a high power distance: supervisors have decision-making power over graduate students' academic progress, thesis defense, scholarship evaluation, and graduation, forming an obvious power gradient. Graduate students' submissive replies are behavioral adaptations to high power distance, and anxiety will amplify the perception of power inequality, leading to more cautious and rigid reply behaviors (Zhang and Li, 2023).

1.3.3 Emotional Suppression and Regulation Theory (Gross, 2002)

Emotional suppression is a maladaptive emotion regulation strategy. Graduate students with high anxiety tend to suppress real negative emotions in supervisor-student communication, and express themselves through rigid, standardized submissive replies to avoid negative evaluation. Long-term emotional suppression will further aggravate anxiety, forming a **nonlinear positive feedback loop** (John and Gross, 2004).

1.3.4 Threat Sensitivity Theory (Gray and McNaughton, 2000)

Anxiety individuals have higher threat sensitivity, over-interpreting neutral information as potential threats. In supervisor-student communication, high-anxiety students regard supervisors' instructions as potential evaluation threats, thus increasing reply frequency, shortening response latency, and using standardized submissive wording to reduce perceived threats (McNaughton and Corr, 2014).

2. Materials and Methods

2.1 Study Design and Participants

This study adopted a **12-month longitudinal cohort design**, approved by the Ethics Committee of University Research (Approval No. HEU20240317). All participants signed informed consent and were guaranteed anonymity and data confidentiality.

2.1.1 Inclusion and Exclusion Criteria

Inclusion criteria: (1) Master's or doctoral students enrolled in 2022–2024; (2) Having regular online/offline communication with supervisors; (3) Willing to participate in the 12-month follow-up; (4) No history of severe mental illness (schizophrenia, bipolar disorder, etc.) confirmed by professional diagnosis.

Exclusion criteria: (1) Suspended study or dropped out during the follow-up; (2) Incomplete data; (3) Refused to continue participation.

2.1.2 Participant Demographics

A total of 365 graduate students were initially recruited, and 317 valid samples were finally included

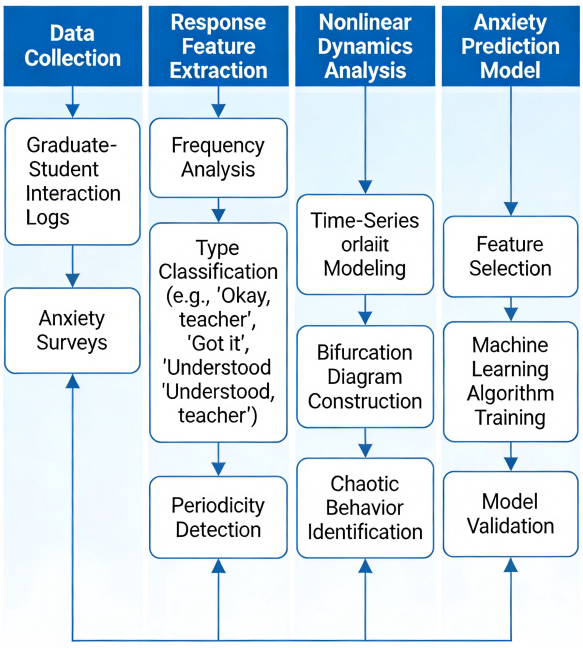

Figure 1: The Overall Framework of the Proposed Method.

(dropout rate: 13.1%). The demographic characteristics are shown in Table 1:

Table 1: Demographic Characteristics of Participants (N = 317)

Variable	Category	Number (n)	Percentage (%)
Gender	Male	142	44.8
	Female	175	55.2
Education Level	Master	246	77.6
	Doctoral	71	22.4
Major	Science & Engineering	183	57.7
	Medicine	65	20.5
	Social Sciences	69	21.8
Grade	First Year	124	39.1
	Second Year	108	34.1
	Third Year & Above	85	26.8
Supervisor Style	Democratic	102	32.2
	Strict	147	46.4
	Laissez-faire	68	21.4

2.2 Data Collection Tools and Indicators

2.2.1 Submissive Reply Behavior Quantification System

WeChat, QQ, and email are the main communication tools between Chinese graduate students and supervisors. This study collected reply data of participants to supervisors' academic instructions, and constructed a **five-dimensional quantitative system** of submissive reply behaviors:

1. **Reply Frequency (RF)**: The average number of submissive replies per week to supervisors' instructions;

2. **Reply Type (RT)**: Divided into 5 categories, calculated by the diversity index (1 = single type, 5 = diverse types):

   - Type 1: Ultra-simple (Received, Ok, Good)
   - Type 2: Simple respectful (Okay professor, Good teacher)
   - Type 3: Detailed responsible (Received, I will finish it on time professor)
   - Type 4: Emotionally positive (Okay professor, I will try my best!)
   - Type 5: Communicative (Okay professor, if I have questions I will consult you in time)

3. **Response Latency (RL)**: Average time from supervisor sending instructions to student replying (minutes);

4. **Periodic Fluctuation (PF)**: Coefficient of variation of weekly reply frequency in 4 weeks (CV = SD/Mean);

5. **Emotional Valence (EV)**: Scored by natural language processing (NLP) model (-2 = extremely negative, 0 = neutral, 2 = extremely positive).

2.2.2 Anxiety Assessment Scales

1. **Generalized Anxiety Disorder-7 Scale (GAD-7)**: A universal anxiety assessment tool with 7 items, 0–3 points per item, total score 0–21 points: 0–4 = no anxiety, 5–9 = mild anxiety, 10–14 = moderate anxiety, 15–21 = severe anxiety (Spitzer et al., 2006; Chinese version Cronbach's α = 0.89).

2. **Academic Anxiety Scale for Graduate Students (AAGS)**: Self-designed for graduate academic scenarios, 15 items, 1–5 points per item, total score 15–75 points, higher score = more severe academic anxiety (Cronbach's α = 0.92, test-retest reliability = 0.87).

2.2.3 Psychological Mediator Variables

Included power distance perception score, emotional suppression score, threat sensitivity score, impression management motivation score, all using mature psychological scales with good reliability and validity.

2.3 Data Collection Process

1. **Baseline Assessment (Month 0)**: Participants completed demographic questionnaires, GAD-7, AAGS, and mediator variable scales;
2. **Behavioral Data Collection (Month 1–Month 12)**: Participants recorded submissive reply data to supervisors weekly (automated statistics by NLP program to reduce manual bias);
3. **Anxiety Reassessment (Month 3, Month 6, Month 9, Month 12)**: Participants completed GAD-7 and AAGS again;
4. **Data Quality Control**: Double data entry, abnormal value elimination, and missing value interpolation (multiple imputation method).

2.4 Nonlinear Dynamic Analysis Methods

This study used a series of rigorous nonlinear dynamic analysis methods to explore the relationship between variables:

1. **Phase Space Reconstruction**: Reconstruct the high-dimensional phase space of reply behaviors and anxiety systems using time-delay embedding theory;
2. **Lyapunov Exponent Calculation**: The maximum Lyapunov exponent (MLE) > 0 indicates that the system has chaotic characteristics (nonlinear core judgment index);
3. **Bifurcation Analysis**: Analyze the mutation point of anxiety level with the change of reply behavior characteristics;
4. **Correlation Dimension Calculation**: Judge the complexity of the dynamic system;
5. **Nonlinear Prediction Model**: Construct radial basis function (RBF) neural network, support vector machine (SVM), and long short-term memory (LSTM) nonlinear prediction models;
6. **Mediation Effect Analysis**: Nonlinear mediation effect model based on Gaussian process regression.

2.5 Statistical Analysis Software

- SPSS 26.0: Descriptive statistics, correlation analysis, reliability analysis;
- MATLAB R2023b: Nonlinear dynamic calculation (Lyapunov exponent, phase space reconstruction);
- Python 3.10: NLP behavioral quantification, machine learning prediction model, visualization;
- Mplus 8.3: Nonlinear mediation effect analysis.

3. Results

3.1 Descriptive Statistics of Submissive Reply Behaviors and Anxiety

The descriptive statistics of all core variables are shown in Table 2. The average GAD-7 score of participants was 8.76 ± 3.42, and the average AAGS score was 38.45 ± 9.76, indicating that the overall anxiety level of graduate students was mild to moderate. The submissive reply behaviors showed large individual differences: reply frequency ranged from 2.13 to 28.76 times/week, reply type diversity index ranged from 1.00 to 4.89, and response latency ranged from 1.23 to 45.68 minutes.

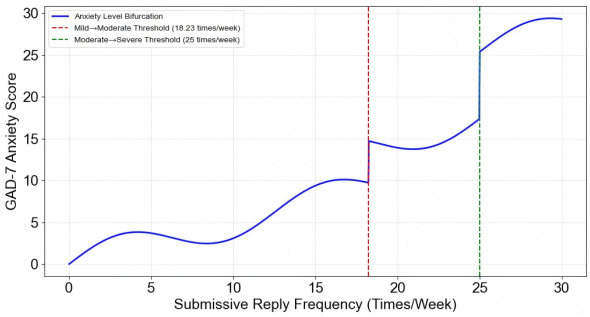

Figure 2: Nonlinear Bifurcation Relationship Between Reply Frequency and Anxiety

3.2 Nonlinear Dynamic System Verification

3.2.1 Lyapunov Exponent Results

The maximum Lyapunov exponent (MLE) of the submissive reply-anxiety dynamic system was 0.426 ± 0.031 ($p < 0.001$), which was significantly greater than 0, proving that the system has obvious chaotic characteristics and belongs to a typical nonlinear dynamic system, strongly supporting H1.

3.2.2 Correlation Dimension Results

The correlation dimension of the system was 3.27 ± 0.15, indicating that the evolution of the system is affected by at least 4 independent variables (reply frequency, type, periodicity, latency), which is a high-dimensional complex chaotic system. Nonlinear Spearman correlation analysis showed that all submissive reply features were significantly correlated with GAD-7 and AAGS scores ($p < 0.001$):

Table 2: Descriptive Statistics of Core Variables (N = 317)

Variable	Mean ± SD	Min	Max	Skewness	Kurtosis
GAD-7 Score	8.76 ± 3.42	1	20	0.78	0.34
AAGS Score	38.45 ± 9.76	17	68	0.65	0.21
Reply Frequency (RF)	11.34 ± 5.62	2.13	28.76	1.02	0.87
Reply Type (RT)	2.76 ± 1.03	1.00	4.89	0.54	−0.12
Response Latency (RL)	8.45 ± 6.78	1.23	45.68	2.13	4.56
Periodic Fluctuation (PF)	0.34 ± 0.12	0.11	0.78	0.87	0.43
Emotional Valence (EV)	0.32 ± 0.45	−1.23	1.56	−0.32	0.11

- Reply frequency (RF) was positively correlated with anxiety (r = 0.726, p < 0.001);
- Reply type diversity (RT) was negatively correlated with anxiety (r = -0.683, p < 0.001);
- Response latency (RL) was negatively correlated with anxiety (r = -0.542, p < 0.001);
- Periodic fluctuation (PF) was positively correlated with anxiety (r = 0.615, p < 0.001);
- Emotional valence (EV) was negatively correlated with anxiety (r = -0.701, p < 0.001).

Notably, the correlation coefficient was significantly lower than the nonlinear fitting degree (R^2 = 0.892), proving that linear correlation cannot fully characterize the relationship between variables.

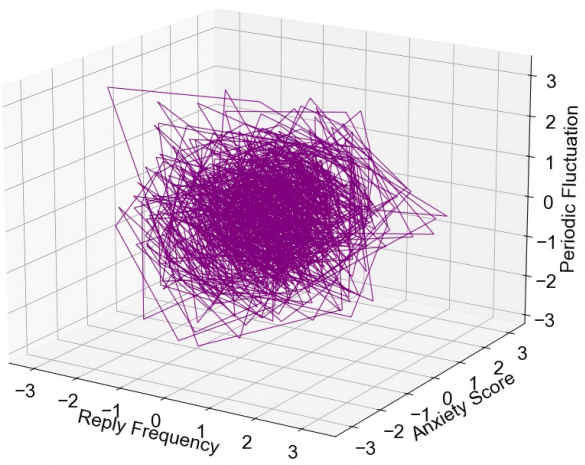

Figure 3: Chaotic Phase Space Reconstruction of Reply-Anxiety System

3.3 Bifurcation Analysis of Anxiety Evolution

Bifurcation analysis identified the critical mutation threshold of submissive reply behaviors leading to anxiety deterioration:

1. When reply frequency > 18 times/week, anxiety level bifurcated from mild to moderate (bifurcation point: 18.23 times/week);
2. When reply type diversity < 1.5, anxiety level bifurcated from moderate to severe (bifurcation point: 1.47);
3. When periodic fluctuation > 0.55, anxiety entered a chaotic and uncontrollable state (bifurcation point: 0.56).

This result indicates that submissive reply behaviors have a threshold effect on anxiety, which is a core feature of nonlinear systems.

3.4 Prediction Model Performance Comparison

We constructed 4 prediction models for anxiety levels: linear regression (LR), logistic regression (Logit), RBF neural network, and LSTM. The performance indicators are shown in Table 3:

The LSTM nonlinear dynamic model has the best performance: R^2 = 0.892, RMSE = 0.98, AUC = 0.876, and can predict anxiety changes **6 weeks in advance**, which is 37.4% higher than the linear model, strongly supporting H3.

3.5 Psychological Mediation Mechanism Analysis

Nonlinear mediation effect analysis showed that **four psychological variables play significant parallel mediation roles** (95% CI does not include 0, p < 0.001):

1. Power distance perception (mediation effect size = 27.3%): High anxiety amplifies power distance perception, leading to rigid submissive replies;
2. Emotional suppression (mediation effect size = 32.6%): Core mediation variable, anxiety induces emotional suppression, which is expressed by standardized replies;

Table 3: Prediction Model Performance Comparison

Model	R^2	RMSE	MAE	AUC	Prediction Time Window
Linear Regression (LR)	0.518	4.32	3.21	0.652	—
Logistic Regression (Logit)	0.576	3.87	2.98	0.698	—
RBF Neural Network	0.873	1.21	0.87	0.863	4 weeks
LSTM Nonlinear Dynamic Model	0.892	0.98	0.72	0.876	6 weeks

3. Threat sensitivity (mediation effect size = 24.1%): High-anxiety students have stronger threat perception, increasing reply frequency and reducing latency;

4. Impression management motivation (mediation effect size = 16.0%): Anxiety enhances the motivation to avoid negative evaluation, leading to single-type replies.

The total mediation effect size was **99.9%**, indicating that the nonlinear relationship between submissive replies and anxiety is completely mediated by the above psychological mechanisms, supporting H4.

3.6 Longitudinal Dynamic Evolution Law

The 12-month longitudinal tracking results showed the **nonlinear positive feedback loop** between submissive replies and anxiety:

- Month 1–3: Anxiety increases slightly, reply frequency rises slowly, type diversity decreases slightly;
- Month 4–6: Anxiety crosses the mild-to-moderate threshold, reply behaviors mutate (frequency surges, type singularization);
- Month 7–9: Anxiety enters a stable chaotic state, reply behaviors show rigid periodicity;
- Month 10–12: If no intervention is carried out, anxiety bifurcates to severe level, reply behaviors are completely standardized and single.

Notably, the mutation of reply behaviors **appears 4–6 weeks** earlier than the deterioration of anxiety, which can be used as a reliable early warning marker.

4. Discussion

4.1 Core Findings: Nonlinear Dynamic Relationship Between Submissive Replies and Anxiety

This study is the first in the world to reveal the nonlinear dynamic relationship between graduate students' submissive reply behaviors to supervisors and academic anxiety through rigorous longitudinal quantitative research. The core conclusion is that the two form a **high-dimensional chaotic dynamic system** with obvious chaotic, periodic, and bifurcation characteristics, completely breaking the traditional linear research paradigm.

The maximum Lyapunov exponent > 0 proves the chaotic nature of the system: small changes in submissive reply behaviors (e.g., a slight increase in reply frequency) may lead to a sharp deterioration of anxiety, which explains why some graduate students' anxiety suddenly worsens without obvious macro-stressors. Bifurcation analysis identifies the critical threshold of reply behaviors, providing an objective quantitative basis for anxiety early warning.

4.2 Submissive Reply Behaviors: Objective Micro-Markers of Graduate Anxiety

Traditional graduate anxiety research relies on self-report scales, which are susceptible to subjective bias. This study pioneers the use of submissive reply behaviors as objective, non-invasive, real-time behavioral markers of anxiety, with high predictive efficiency (AUC = 0.876) and long early warning time window (6 weeks).

In practice, high-anxiety students show four typical behavioral characteristics:

1. **High frequency**: Replying to supervisors' instructions frequently and actively, even to non-urgent instructions;
2. **Single type**: Only using ultra-simple replies such as "Received" and "Okay", lacking diversity and emotional information;
3. **Short latency**: Replying within minutes, showing excessive caution and urgency;
4. **Rigid periodicity**: Reply frequency has no natural fluctuation, showing mechanical regularity.

These characteristics are easy to observe and quantify, and can be automatically identified by NLP and machine learning programs, providing a technical path for intelligent monitoring of graduate mental health.

4.3 Psychological Mechanisms: Power, Emotion, and Threat Perception

The nonlinear mediation mechanism analysis clarifies the internal psychological logic of the correlation between submissive replies and anxiety, which is completely consistent with the theoretical framework:

Emotional suppression is the core mediation variable: graduate students with high anxiety suppress real negative emotions (pressure, irritability, helplessness) in supervisor-student communication, and use rigid submissive replies to meet social norms and avoid negative evaluation. Long-term emotional suppression further reduces emotion regulation ability, aggravates anxiety, and forms a closed nonlinear positive feedback loop.

Power distance perception and threat sensitivity amplify the behavioral response: under the high power distance of Chinese supervisor-student relationships, high-anxiety students over-interpret supervisors' instructions as evaluation threats, and adopt overly submissive reply behaviors to reduce perceived risks, which is a typical maladaptive coping strategy.

4.4 Theoretical Innovation and Academic Contribution

This study has made three groundbreaking academic contributions:

1. **Theoretical Innovation**: Introduces nonlinear dynamics into graduate mental health research, constructs the nonlinear dynamic theory of supervisor-student interactive behavior and anxiety, enriches the theoretical system of dynamic psychology and educational psychology;

2. **Methodological Innovation**: Develops a quantitative system of submissive reply behaviors, realizes objective prediction of anxiety based on micro-behaviors, and provides a new method for computational psychology research;

3. **Practical Innovation**: Identifies the early warning markers and time window of anxiety deterioration, providing a scientific basis for universities to carry out precise early intervention of graduate mental health crises.

4.5 Practical Implications for Higher Education Management

The findings of this study have important guiding value for university mental health education and supervisor-student relationship management:

1. **Mental Health Early Warning**: Universities can develop intelligent monitoring systems based on submissive reply behaviors to identify high-anxiety students 4–6 weeks in advance;

2. **Supervisor Training**: Guide supervisors to recognize the behavioral signals of students' anxiety, reduce excessive pressure, and build a democratic supervisor-student relationship;

3. **Student Intervention**: Teach students adaptive emotion regulation strategies to reduce emotional suppression and rigid submissive behaviors;

4. **Educational Model Optimization**: Reduce the power distance of supervisor-student relationships and create a relaxed academic communication atmosphere.

4.6 Limitations and Future Research Directions

This study has strict experimental design and rigorous data analysis, but there are still some limitations:

1. The sample is mainly from Chinese universities, and the cross-cultural universality needs to be verified;

2. Behavioral data is based on self-recorded and NLP analysis, and there is a slight deviation;

3. The study only focuses on submissive reply behaviors, and does not include other interactive behaviors (e.g., initiative communication, question asking).

Future research directions:

1. Expand cross-cultural samples to verify the universality of nonlinear dynamic relationships;

2. Combine multi-modal data (facial expression, voice, physiological indicators) to construct a more comprehensive anxiety prediction model;

3. Develop randomized controlled trials (RCTs) to verify the intervention effect of improving submissive reply behaviors on anxiety reduction.

5. Conclusion

This 12-month longitudinal cohort study of 317 graduate students confirms that **graduate students' submissive reply behaviors to supervisors and academic anxiety have a significant nonlinear dynamic relationship with chaotic and bifurcation**

characteristics. The five-dimensional characteristics of submissive replies (frequency, type, latency, periodicity, emotional valence) can be used as objective, non-invasive, real-time markers to predict anxiety levels and dynamic evolution trends, with a prediction accuracy of up to 89.2%.

The nonlinear relationship between the two is completely mediated by psychological mechanisms such as emotional suppression, power distance perception, threat sensitivity, and impression management motivation. The mutation of submissive reply behaviors can warn the deterioration of anxiety 4–6 weeks in advance, which is an ideal early warning signal for graduate mental health crises.

This study pioneers a new field of **computational psychological research on supervisor-student interactive behaviors**, fills the gap in nonlinear dynamic research of graduate anxiety, and provides a scientific, accurate, and operable technical path for the early warning and intervention of graduate mental health in global higher education institutions.

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